

18 INSTALLATION AND OPERATING PROCEDURES

This section briefly describes the basic installation procedure for MARS 3.1 and the program execution steps. The platforms supported for the current (and most probably for the future) release of MARS are MS Windows O/S systems, i.e., Windows 95, Windows 98, Windows NT, and Windows 2000, all of them running on PC. A version for the Linux operating system is under development.

18.1 MARS 3.1 Code Installation

The files released with MARS 3.1 and their functions are:

1. mars31.exe: MARS 3.1 main program
2. contl.dll: Windows DLL for CONTEMPT4 coupled run
3. tpfh20: Steam table data for light water
4. tpfd20: Steam table data for heavy water
5. grfx.exe: post-plot data processor of 3-D grafout file

MARS 3.1 can be installed by simply following the procedure below.

1. Copy all the distributed files in any of the directory preferred. (eg. C:\MARS3.1)
2. Create a shortcut for mars31.exe file on the desktop.

18.2 MARS 3.1 Code Execution Steps

Once MARS 3.1 has been installed and input data file is ready, proceed as follow to execute the program. It is assumed that MARS 3.1 has been installed in C:\MARS3.1 and user's input files reside in D:\Project_1\Case_1.

1. User may create the input data file in whichever directory location preferred. However, when the directory other than MARS 3.1 main directory (C:\MARS3.1 in our example) is used as a working directory, steam table file should be also copied from the MARS 3.1 main directory. The working directory also needs contl.dll if MARS3.0/CONTEMPT4 coupled analysis is performed. The contain.dll or master.dll is necessary for the similar coupled analysis, such as containment analysis or 3-D kinetics calculations.

2. Click the right mouse button on the MARS 3.1 shortcut, and then choose registration button. You will see the dialog window similar to **Fig. 18-1**. Specify your MARS 3.1 working directory (eg. d:\Project_1\Case_1) as a 'Starting directory' and close the window by clicking OK. This step needs to be performed only once, unless any other directory is used as a working directory.

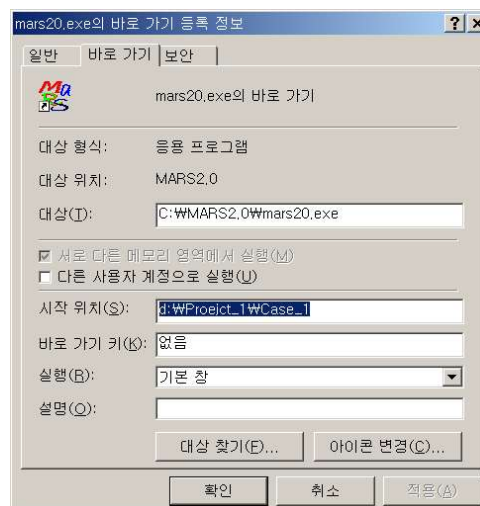


Figure 18-1. A View of MARS Execution Setting Dialog Window

3. Now, double click the MARS 3.1 shortcut on your desktop. Three new windows as shown in **Fig. 18-2** will appear. In the top dialog window you can specify input and output data file names if the default file names provided are not appropriate. When 3-D module is used, one can also specify i/o data file names for 3-D analysis by clicking 'More 3D' button. Once the data file names are assigned properly, you are ready to the start analysis run by clicking 'Start Run' button. But if the output files are already present (if it is not the very first run in this directory), the program terminates by file opening error. Users, therefore, have to remove the previously generated output files by clicking 'Delete Files' button before the execution. The 'delete files' button pops up another window where you can select the problem type. For instance, if 'new' problem type is selected, all the file generated in the previous run will be deleted. The 'rstplt' file could be remained undeleted if 'restart' button is clicked on. The window provides file delete functions for coupled analysis as well.

4. Once analysis run starts, you may want maximize the MARS 3.1 main window by clicking max button on the right upper corner of the window. Also rearranging the graph windows and the text info window to 'tile' form will enhance your interaction with several windows (See **Fig. 18-3**). The graph windows show the minor edit variables specified in the input file, and the text info box displays various information as the run proceeds.

Beside above procedure, data file named 'inputc' should exist when CONTEMP4 module is invoked by the input data. The input file name is currently fixed as 'inputc' and the dialog box does not allow for any change. In running the MARS/CONTEMP4 coupled problem, users should manually remove the CONTEMP4 output files before the execution. Refer to CONTEMP4 manual for the preparation of 'inputc'.

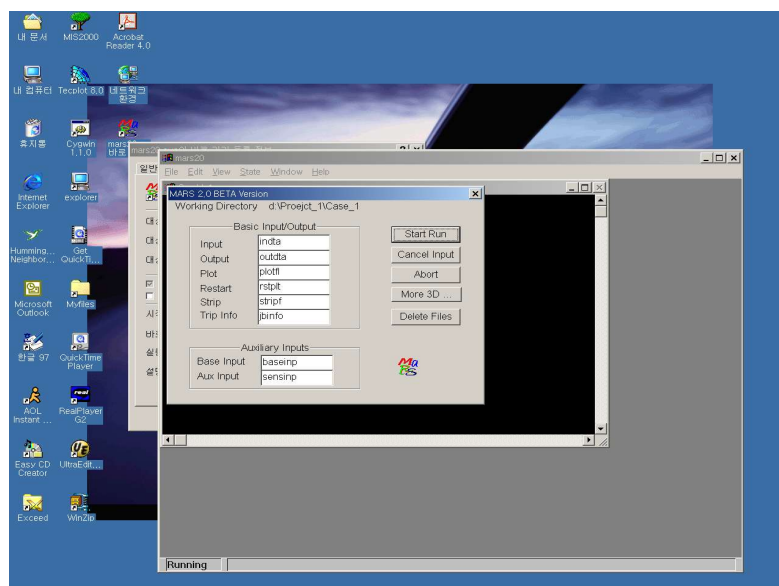


Figure 18-2. A View of Dialog Box Listing the Input and Output Files to Use

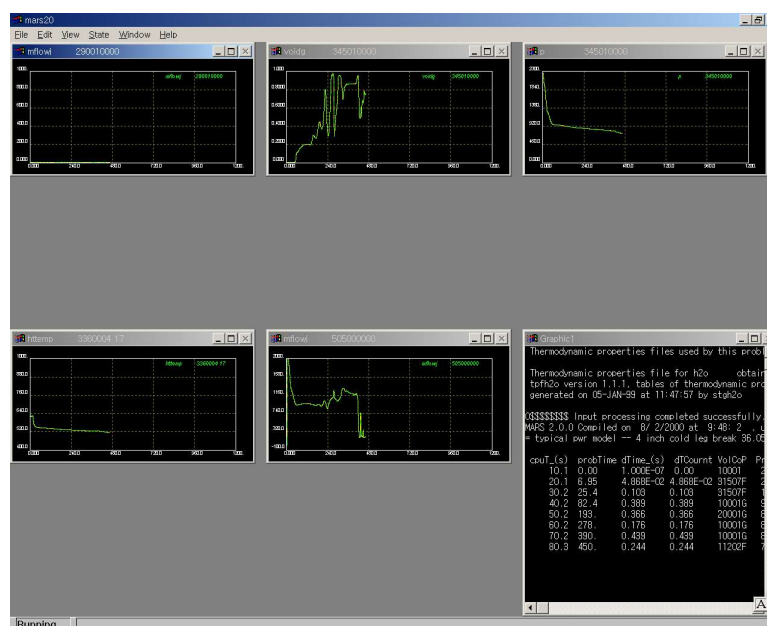


Figure 18-3. A View of MARS Running in Qwin Mode.

18.3 MARS 3.1 I/O Files

The input and output file names provided by default are as below. These names can be overridden by the user using the dialog box, as mentioned in Section 18.2.

Files related with both 1-D and 3-D Vessel modules

indta	input data
plotfl	minor edit data for plot
jbinfo	trip information data
coupfl	contour plot data

Files related with 1-D module only

outdta	printed output
rstplt	restart plot file
stripf	the strip file
screen	text sub-window output

Files related with 3-D module only

restart	restart file
grafout	plot data (needed for restart run)
mserr	mass error data
output	printed out for
mvyout	text sub-window output

Files related with CONTEMPT4 module

inputc	input file for CONTEMPT4 module
ouputc	text output file
plotflc	plot data file

screen text sub-window output

Files related with CONTAIN2.0 module

inputc input file for CONTAIN module

ouputc text output file

plotflc plot data file

Files related with 3D kinetics MASTER module

MAS_INP input file for MASTER Module

Note: Other I/O files for MASTER Modules are specified in 'MAS_INP'.

Files related with MIDAS connection

Trip4Midas.dat trip data file

kinetic4Midas.dat reactor kinetics data file

VolJun4Midas.dat volume/junction/component data file

Heat4Midas.dat heat structure data file

MWR4Midas.dat metal-water reaction data file

reflood4Midas.dat reflood data file

control4Midas.dat control variable data file



APPENDIX A: "MAS_MAP" Input Data Generation

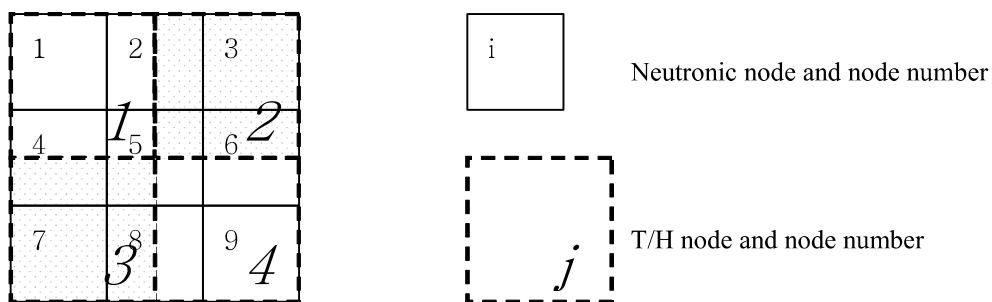
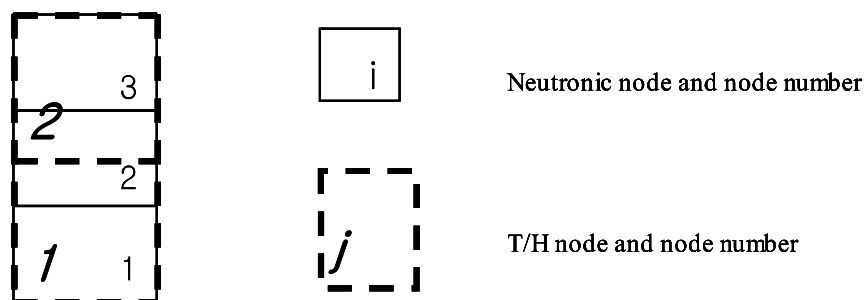
In general, the mesh structures used for the neutronic and thermal/hydraulic (T/H) calculations are different from each other. In such cases, the correspondence between the neutronic and T/H nodes is no longer one-to-one and a special input is required to define the correspondence. In the MARS/MASTER coupled code, the correspondence is established through the MAS_MAP file in which the schemes of mapping a neutronic node onto T/H nodes are specified. It is assumed in the mapping input scheme described by MAS_MAP that the same radial node structure is applied in all axial levels. Due to this assumption, it is not necessary to provide the mapping information for every neutronic node in the three-dimensional domain. Instead, it is sufficient to specify only the radial and the axial mapping information separately.

The MAS_MAP file consists of three sets of data with each set starting with a comment card. The first set has nothing to do with the mapping, but it defines the fuel surface temperature weighting factor, ω , that is used to obtain the effective fuel temperature out of the fuel centerline (T_f^{CL}) and surface (T_f^S) temperatures as:

$$T_f^{Dop} = \omega T_f^S + (1 - \omega) T_f^{CL}.$$

The second and third sets provide the radial and axial mapping information, respectively. The same rule to define the mapping scheme described below applies to both directions. Each line in the mapping input represents a neutronic node, either radially or axially. The line starts with a neutronic node number and lists pairs of two numbers consisting of a heat structure (fuel rod) number in the T/H input and the volume fraction of the neutronic node belonging to the heat structure. Note that a neutronic node can belong to only one heat structure node or different parts of the node can belong to different heat structure nodes. The sum of the fractions of a neutronic node belonging to different heat structure nodes must 1.0. The radial mapping information must be given from the first radial node to the last. The same rules are applied to the axial mapping information as well.

Consider a reactor core, of which neutronics is modeled by 9 radial meshes and 3 vertical meshes (See **Figure A1**) and of which T/Hs is modeled by 4 radial meshes and 2 vertical meshes (See **Figure A2**). A sample MAS_MAP file is given in **Figure A3** for the neutronic-to-T/H mapping scheme.

**Figure A1.** Radial Mapping Example**Figure A2.** Axial Mapping Example

```

pellet surface temp weighting factor
0.7
radial mapping (neutnode, (thnode(i),frac(i)),i=1,nth))
1      1  1.0000
2      1  0.5000  2  0.5000
3      2  1.0000
4      1  0.5000  3  0.5000
5      1  0.2500  2  0.2500  3  0.2500  4  0.2500
6      2  0.5000  4  0.5000
7      3  1.0000
8      3  0.5000  4  0.5000
9      4  1.0000
axial mapping (neutnode, (thnode(i),frac(i)),i=1,nth)) 3 neutronic and 2 t/h planes
1      1  1.0000
2      1  0.5000  2  0.5000
3      2  1.0000

```

Figure A3. Sample MAS_MAP file